Q.8)

	Allocation			Need			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	7	4	3	3	3	2
P_1	2	0	0	1	2	2			
P_2	3	0	2	6	0	0			
P_3	2	1	1	0	1	1			
P_4	0	0	2	4	3	1			

Work vector = 3, 3, 2

Need (P_0) =/< work. Not executed

Need (P_1) <= work. executed

work vector = 5, 3, 2

Need (P_2) <= work

Need (P_3) <= work. Executed

work Vector = 7, 5, 5

Need (P_2) <= work

work vector = 10, 5, 7

a) The system is in safe state

b) Safe sequence is $P_1 \rightarrow P_3 \rightarrow P_4 \rightarrow P_0 \rightarrow P_2$

c) If need (P_1) = 3, 0, 2

There will be no change, the system is still in safe state as work vector = 3, 3, 2.

d) Yes

e) Yes

Q.9) Work Vector = 0, 0, 1

Need (P_0) <= Work, P_0 executed

Work = 0, 1, 1

Need P_2 <= Work, P_2 is executed

Work = 3, 1, 3

Need P_3 <= work

Work = 5, 2, 4

Need P_4 <= work

Work = 5, 2, 5

Need P_1 <= work

Work = 7, 2, 6

(i) Not in deadlock ; Safe Sequence → $P_0 \to P_2 \to P_3 \to P_4 \to P_1$

(ii) No effect. Still in Safe State.

Q.10)

	Allocation				Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	0	0	1	5	2	0
P_1	1	0	0	0	0	7	5	0				
P_2	1	3	5	4	1	0	0	2				
P_3	0	6	3	2	0	0	2	0				
P_4	0	0	1	4	0	6	4	2				

Need (P_0) <= work. P_0 = True

Work Vector = 1, 5, 3, 2

Need (P_1) <= Work

Need (P_2) <= work

Work = 2, 8, 8, 6

Need (P_3) <= Work

Work = 2, 14, 11, 8

Need (P_4) <= work

Work = 2, 14, 12, 12

Need P_1 <= work

Work = 3, 14, 12, 12

2) Safe Sequence → $P_0 \to P_2 \to P_3 \to P_4 \to P_1$

3) Yes, it can be granted immediately as work vector

⇒ 1, 5, 2, 0

Q.11)

	Allocation			Need			Available		
	R_1	R_2	R_3	R_1	R_2	R_3	R_1	R_2	R_3
P_1	2	2	3	1	4	5	7	7	10
P_2	2	0	3	2	3	0			
P_3	1	2	4	2	2	0			

Work = 7, 7, 10

Need P_1 < work

Work = 9, 9, 13

Need P_2 < work

Work = 11, 9, 16

Need P_3 < work

Work = 12, 11, 20

(i) Safe Sequence : $P_1 \rightarrow P_2 \rightarrow P_3$

(ii) a) Yes

b) yes

c) yes

Q.12) Work Vector = (2, 1, 0, 0)

Need (P_0) <= Work

Need (P_2) <= work. P_2 = true

Work = 2, 2, 2, 0

Need (P_0) <= work

Need (P_1) <= work

Work = 4, 2, 2, 1

Need (P_0) <= work. work = 4, 2, 3, 1

Safe Sequence: $P_2 \rightarrow P_1 \rightarrow P_0$

Q.13)

100	500	200	300	600

1) Best Fit for 212, 417, 112, 426

212 → 300

100	500	200	88	600

417 → 500

100	83	200	88	600

112 → 200

100	83	88	88	600

426 → 600

100	83	88	88	174

2) First Fit

212 → 500

100	288	200	300	600

417 → 600

100	288	200	300	183

112 → 288

100	176	200	300	183

426 → Not enough space

3) Worst Fit

212 → 600

100	500	200	300	388

417 → 500

100	83	200	300	388

112 → 388

100	83	200	300	276

426 → Not enough space

Q. 14) 1000K Memory

Best fit :

$J_1 \rightarrow 100$ | 200 | 800 |

$J_2 \rightarrow 350$ | 200 | 350 | 450 |

$J_3 \rightarrow 300$ | 200 | 350 | 300 | 150 |

First fit allocation :-

Step	Memory State
J1 (200)	Free = 1000 [J1] [800 free]
J2 (350)	[J1] [J2] [450 free]
J3 (300)	[J1] [J2] [J3] [150 free]
J1 finishes	[200 free] [J2] [J3] [150 free]
J4 (120)	fits in 200 → [J4] [80 free] [J2] [J3] [150 free]
J5 (150)	fits in 150 → [J4] [80 free] [J2] [J3] [J5]
J6 (80)	fits in 80 → [J4] [J6] [J2] [J3] [J5]

Best fit allocation :-

After J1 finishes	[200 free] [J2] [J3] [150 free]
J4 (120)	goes to 150 → leaves 30
J5 (150)	goes to 200 → leaves 50
J6 (80)	cannot fit in 50 or 30

↳ External fragmentation increases

Q. 15) Heap :

block	Size
Job 1	50 K
Job 2	300 K
free	150 K
Job 3	600 K
free	350 K

Requests : 300, 25, 125, 50

* **Analysis** :

300 → fits exactly in 350
25 → fits in 50 or 150
125 → fits best in 150
50 → remaing depends

Q.16) **2 Frames**

FIFO

	1	1	2	2	1	4	2	3	3	5	5	4
			2	2	2	2	2	3	3	3	3	4
	1	1	1	1	1	4	4	4	4	5	5	5
	F	H	F	H	H	F	H	F	H	F	H	F

No of faults = 6

LRU

	1	1	2	2	1	4	2	3	3	5	5	4
			2	2	2	4	4	3	3	3	3	4
	1	1	1	1	1	1	2	2	2	5	5	5
	F	H	F	H	H	F	F	F	H	F	H	F

no of faults = 7

OPTIMAL

	1	1	2	2	1	4	2	3	3	5	5	4
			2	2	2	2	2	3	3	5	5	5
	1	1	1	1	1	4	4	4	4	4	4	4
	F	H	F	H	H	F	H	F	H	F	H	H

no of fault = 5

3 Frames

FIFO

	1	1	2	2	1	4	2	3	3	5	5	4
						4	4	4	4	4	4	4
			2	2	2	2	2	2	2	5	5	5
	1	1	1	1	1	1	1	3	3	3	3	3
	F	H	F	H	H	F	H	F	H	F	H	H

no of faults = 5

LRU

	1	1	2	2								
						4	4	4	4	5	5	5
	1	1	2	2	2	2	2	2	2	2	2	4
	1	1	1	1	1	1	1	3	3	3	3	3
	F	H	F	H	H	F	H	F	H	F	H	F

OPTIMAL

1	1	2	2	1	4	2	3	3	5	5	4
					4	4	4	4	4	4	4
		2	2	2	2	2	2	2	5	5	5
1	1	1	1	1	1	1	3	3	3	3	3
F	H	F	H	H	F	H	F	H	F	H	H

no of faults = 5

4 Frames : FIFO

1	1	2	2	1	4	2	3	3	5	5	4
					4	4	4	4	4	4	4
		2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	5	5	5
F	H	F	H	H	F	H	F	H	F	H	H

no of faults = 5

LRU:

1	1	2	2	1	4	2	3	3	5	5	4
							3	3	3	3	3
					4	4	4	4	4	4	4
		2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	5	5	5

No of faults = 5

OPTIMAL:

1	1	2	2	1	4	2	3	3	5	5	4
							3	3	3	3	3
					4	4	4	4	4	4	4
		2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	5	5	5
F	H	F	H	H	F	H	F	H	F	H	H

No of faults = 5

Q.17) 5 frames

FIFO

1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
				5	5	5	5	5	5	5	5	7	7	7	7	7	7	7	7
			4	4	4	4	4	4	4	4	3	3	3	3	3	3	3	3	3
		3	3	3	3	3	3	3	3	2	2	2	2	2	2	2	2	2	2
	2	2	2	2	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6	6

No. of faults = 10

LRU

1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
						5	5	5	5	5	7	7	7	7	7	7	7	7	7
			4	4	4	4	4	4	4	4	3	3	3	3	3	3	3	3	3
		3	3	3	3	3	6	6	6	6	6	6	6	6	6	6	6	6	6
	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

No. of faults = 8

OPTIMAL

1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
						5	5	5	5	5	5	7	7	7	7	7	7	7	7
			4	4	4	4	6	6	6	6	6	6	6	6	6	6	6	6	6
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

No. of faults = 7

4 frames

1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
			4	4	4	4	4	4	1	1	1	1	1	1	2	2	2	2	2
		3	3	3	3	3	3	2	2	2	2	2	2	6	6	6	6	6	6
	2	2	2	2	2	2	6	6	6	6	6	7	7	7	7	7	7	3	3
1	1	1	1	1	1	5	5	5	5	5	3	3	3	3	3	1	1	1	1

Faults = 14

1 mk.

LRU

1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
			4	4	4	4	6	6	6	6	6	7	7	7	7	1	1	1	1
		3	3	3	3	5	5	5	5	5	3	3	3	3	3	3	3	3	3
	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	6	6	6	6

No. of faults : 10

OPTIMAL

1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
			4	4	4	5	6	6	6	6	6	6	6	6	6	6	6	6	6
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	1	7	7	7	7	1	1	1	1

No. of faults = 8

Q.18) Requests : 98, 183, 37, 122, 14, 124, 65, 67
Initial head = 53

Total Head Movement :

Algorithm	Movement
FCFS	640
SSTF	236
SCAN	208
C-SCAN	322
LOOK	208
C-LOOK	326

SSTF & LOOK have minimal movement :-

53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183

12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 → 236

LOOK:

SPLIT: left of 53: 37, 14

Right of 53: 65, 67, 98, 122, 124, 183

So, 53 → 37 → 14 → 65 → 67 → 98 → 122 → 124 → 183

↓

16 + 23 + 51 + 2 + 31 + 24 + 2 + 59 = $\boxed{208}$

Q.19 Queue: 86, 147, 91, 177, 94, 150

Initial: 143

FCFS: 143 → 86 → 147 → 91 → 177 → 94 → 150

→ 57 + 61 + 56 + 86 + 83 + 56 = 399

SSTF: 143 → 147 → 150 → 177 → 94 → 91 → 86

→ 4 + 3 + 27 + 83 + 3 + 5 = 125

C-SCAN: ~~143 → 147 +~~ 143 → 147 → 150 → 177 → 199 → 0 → 86 → 91 → 94

→ 4 + 3 + 27 + 22 + 199 + 86 + 5 + 3 = 349

Q.20) Queue: 86, 1470, 913, 1774, 948, 1750, 130

Initial: 143

FCFS: 143 → 86 → 1470 → 913 → 1774 → 968 → 1750 → 130

⇒ 57 + 1384 + 557 + 861 + 826 + 802 + 1620 ⇒ $\boxed{6107}$

SSTF: Order: 143 → 130 → 86 → 913 → 948 → 1470 → 1750 → 1774

= 13 + 44 + 827 + 35 + 522 + 280 + 24

$\boxed{= 1754}$

SCAN: 143 → 913 → 948 → 1470 → 1774 → 4999 → 130 → 86

= 770 + 35 + 522 + 280 + 24 + 3225 + 4869 + 44

$\boxed{= 9769}$

C-SCAN: 143 → 913 → 948 → 1470 → 1750 → 1774 → 4999 → 0 → 86 → 1

= 770 + 35 + 522 + 280 + 24 + 3225 + 4999 + 86 + 244

$\boxed{= 9985}$

LOOK: 143 → 913 → 948 → 1470 → 1750 → 1774 → 130 → 86

→ 770 + 35 + 522 + 280 + 24 + 1644 + 44 = $\boxed{3319}$

C-LOOK: 143 → 913 → 948 → 1470 → 1750 → 86 → 130

→ 770 + 35 + 522 + 280 + 24 + 1688 + 44

→ $\boxed{3363}$